

MINOR PROJECT REPORT
ON
DATA ENCRYPTION STANDARD (DES)

Submitted in partial fulfillment of the requirements for the award of degree

Bachelor of Technology (Computer Science)

Submitted By:
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Roll Number

Under the Guidance of:
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Academic Year: 2025–26

CERTIFICATE

This is to certify that the minor project report entitled “**Data Encryption Standard (DES)**” submitted by the student in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science is a record of bonafide work carried out under my guidance and supervision.

Guide Signature
Name & Designation

Head of Department
College Name

ACKNOWLEDGEMENT

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ABSTRACT

Data Encryption Standard (DES) is a symmetric key encryption algorithm that uses a single secret key for both encryption and decryption. This project explains the working principle, architecture, cryptographic operations, key generation, security analysis, and applications of DES. The aim of this report is to provide a clear and structured understanding of DES for academic purposes.

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CHAPTER 1: INTRODUCTION TO CRYPTOGRAPHY AND DES

This chapter explains the concepts related to the topic in detail. Theoretical discussion, diagrams, and explanations are included to ensure clarity and academic completeness of the Data Encryption Standard.

CHAPTER 2: BACKGROUND AND FUNDAMENTALS OF DES

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CHAPTER 3: DES ARCHITECTURE AND CRYPTOGRAPHIC OPERATIONS

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CHAPTER 4: KEY GENERATION AND KEY MANAGEMENT

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CHAPTER 5: ENCRYPTION AND DECRYPTION PROCESS

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CHAPTER 6: MODES OF OPERATION AND SECURITY ANALYSIS

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CHAPTER 7: VARIANTS, APPLICATIONS AND CONCLUSION

This chapter explains the concepts related to the topic in detail. Theoretical discussion, diagrams, and explanations are included to ensure clarity and academic completeness of the Data Encryption Standard.

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